

11	(a) (i) loss of (signal) power	B1	[1]
	(ii) unwanted power (on signal) that is random	M1 A1	[2]
	(b) for digital, only the 'high' and the 'low' / 1 and 0 are necessary variation between 'highs' and 'lows' caused by noise not required	M1 A1	[2]
	(c) attenuation = $10 \lg(P_2 / P_1)$ <i>either</i> $195 = 10 \lg(\{2.4 \times 10^3\} / P)$ <i>or</i> $-195 = 10 \lg(P / 2.4 \times 10^3)$ $P = 7.6 \times 10^{-17} \text{ W}$	C1 C1 A1	[3]

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